

Mechanical Part

⚠ Note: This robot is a research platform, and rapid improvements are constantly being made. While most of the information provided is accurate, we recommend double-checking for the most up-to-date details. For further inquiries, feel free to contact us at rby.support@rainbow-robotics.com or discuss on our GitHub Discussions page at [rby1-sdk Discussions](#).

Motor Specifications

The table provides details of each motor's gear type, motor power, maximum measurable current, speed, and range of motion. For example, the R/LWH model has a planetary gear system and uses a 200W motor. Other systems utilize harmonic gears, and the movement range and maximum speed vary according to each axis.

	Gear Type	Motor Power	Max. measurable current/ Limit current	Max Speed	Range
R/LWH	Planetary	200W	150 RPM (1.5m/s)	Infinite	
AR	Harmonic	250(450)W	53/46 A	120°/s	-20° ~ 20°
AP	Harmonic	250(450)W	53/46 A	120°/s	-60° ~ 70°
KN	Harmonic	250(450)W	53/46 A	120°/s	-140° ~ 45°
HP	Harmonic	200(280)W	26/23 A	180°/s	-45° ~ 90°
HR	Harmonic	200(280)W	26/23 A	180°/s	-30° ~ 30°

	Gear Type	Motor Power	Max. measurable current/ Limit current	Max Speed	Range
HY	Harmonic	200(280)W	26/23 A	180°/s	-90° ~ 90°
R/LSP	Harmonic	80(180)W	26/14 A	180°/s	-135° ~ 135°
R/LSR	Harmonic	80(180)W	26/14 A	180°/s	R : -180° ~ 4° L : -4° ~ 180°
R/LSY	Harmonic	80(180)W	26/14 A	180°/s	-120° ~ 120°
R/LELB	Harmonic	50(75)W	8/7 A	180°/s	-150° ~ 0°
R/LWY1	Harmonic	30(95)W	8/7 A	360°/s	-360° ~ 360°
R/LWP	Harmonic	30(95)W	8/7 A	360°/s	-100° ~ 115°
R/LWY2	Harmonic + Pulley	45W	8/7 A	360°/s	-170° ~ 170°

Torque Specifications

This table details the torque performance of various motor types. It includes continuous torque, repeated peak torque, and momentary peak torque in Newton-meters (Nm). Additionally, the table specifies the gear ratio, output speed in degrees per second (deg/s), motor encoder resolution in bits, transmission error in arcseconds, stiffness in Newton-meters per radian (Nm/rad), and the motor torque constant in Newton-meters per Ampere (Nm/A).

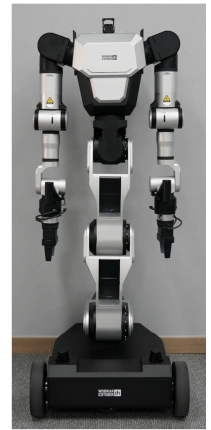
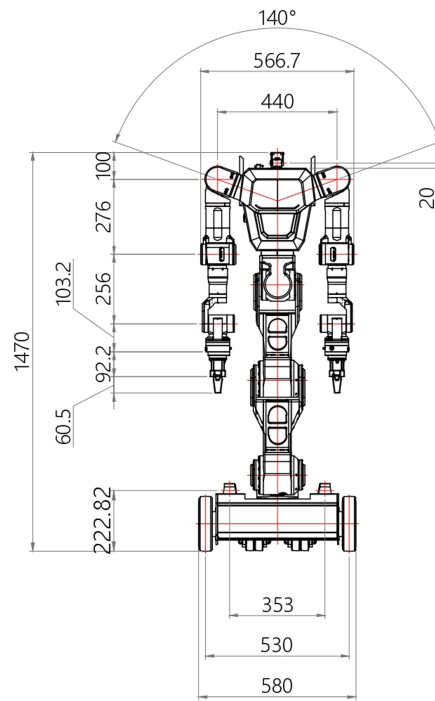
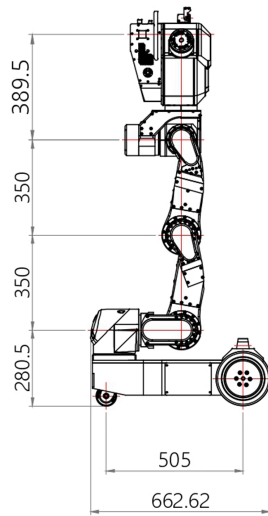
Type	AR, AP, KN	HP, HR, HY	R/L SP, R/L SR, R/L SY	R/L ELB	R/L WY1, R/L WP, R/L WY2
Continuous Torque (Nm)	250	125	65	35	14

Type	AR, AP, KN	HP, HR, HY	R/L SP, R/L SR, R/L SY	R/L ELB	R/L WY1, R/L WP, R/L WY2
Repeated Peak Torque (Nm)	433	204	70	36	36
Momentary Peak Torque (Nm)	841	369	143	70	70
Output Speed (deg/s)	120	180	180	180	360
Motor Encoder Resolution (bit)	14	14	14	14	14
Transmission Error (arcsec)	~50	~50	~50	~50	~50
Stiffness (Nm/rad)	110,000	50,000	14,000	6,100	6,100
Motor Torque Constant (Nm/A)	0.15	0.10	0.10	0.10	0.05

Robot Dimensions

The diagrams show the front and side views of the robot with detailed measurements. The total height of the robot is 1470 mm, and the width is 566.67 mm, with a 140-degree range of motion. The base dimensions are 662.62 mm in length and 580 mm in width. These measurements provide an understanding of the physical space the robot occupies, as well as its potential reach and movement range.

 **Note:** Physical limits and control limits may differ. For detailed operation ranges, please refer to the [URDF files](#).



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